Resolution Geometry: Cornering the Floating Box

Protocolism and Realism, Einstein and Newton, Luck and God

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Abstract

Resolution Geometry (RG) cannot derive general relativity. The source remainder leaves the Weyl curvature free, so the field equation stays a selection or a measurement. What RG can do is *corner* a claim: prove the exact protocol region in which the claim is decided, the region in which it is observationally idle, and the declarations it needs. This paper corners five sets of claims, each with a computed witness.

(1) **The floating box.** Einstein's equivalence principle is a *resolution window*. The connection is removable at a point, the curvature is not. An orbiting gradiometer resolves the tidal tensor at signal-to-noise $\sim 10^{5-6}$. Even an ideal box reads only the energy footprint: the tidal part R_{0i0j} is free of γ, while the spatial part is proportional to γ. The box is therefore blind to half of light bending: Einstein 1911 gives $0.876''$, general relativity 1915 gives $1.751''$.

(2) **Protocolism against realism.** Every observable task factors through the observable quotient, so the latent surplus of any realism is idle, though not refuted. But protocolism is not self-founding: it needs a declared task and declared units.

(3) **RG against Einstein.** General relativity wins every empirical test RG can run. RG corners its ontology:

- the field equation is selected, not forced;
- manifold points are gauge (the hole argument);
- interiors and topology are remainders;
- G and M appear only as GM.

Einstein's *local* realism in quantum mechanics loses; his determinism survives only nonlocally.

(4) **RG against Newton.** Absolute velocity lies in the kernel, so Leibniz wins there. Absolute rotation lies in the quotient, so Newton wins the bucket. Newton's single potential is exact for flat structure at any field strength and fails exactly where structure is resolved.

(5) **Luck and God.**

- Luck is certifiable only *relative to a declared class of gamblers*. Across 37.7 Mbit, the deterministic coin, $\sqrt{2}$ and masked streams beat every finite-state gambler, and every one of them loses 7.3×10^5 nats to its committed oracle.
- Physical luck needs a Bell test plus eight named gates. In 90 million synthetic Bell rounds, countermodels pass the statistics while breaking exactly one gate each, so the physical certificate is *not issued*.
- Ontological luck is not definable by any protocol. Luck in the kernel is never forgotten.
- A God-observer faces a dilemma. As a protocol it must be contextual (0 of 512 Peres–Mermin assignments), information-limited (one bit per step of chaotic horizon) and unit-conventional. As a non-protocol it lies outside the quotient, and RG is silent.

There is no overall winner in the metaphysical sense. RG holds the upper hand on *what observation decides*. General relativity holds it on *what the physics is*, within its domain. No one wins *what exists*. Battery: 18 witnesses, F01–F18, of which F15 and F18 import the final luck battery; 10 computed canaries rejected; master seed 20261026.

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1 What it means to corner a claim

Definition 1.1 (Cornering)**.** Let C be a claim and $\mathcal{P}$ a declared class of protocols. RG *corners* C when it proves three things:

(i) the subclass of $\mathcal{P}$ in which C is decided, by a LOCK or a refutation;

(ii) the subclass in which C lies in a single observational fibre with its rivals (CORRESPONDENCE);

(iii) the declarations that no protocol supplies but C needs.

A party holds the *upper hand* in an arena when its claim survives (i) with the smaller declaration budget (iii). A claim confined entirely to (ii) is *idle*, which is not the same as *refuted*.

Cornering is weaker than refutation and stronger than doubt. It is the only weapon RG has against general relativity, because the physical path paper proved that no function of the source remainder fixes the gravitational field [1]. Every corner below is backed by a computed witness (Section 10). Imported results are marked.

2 The floating box

Einstein's "happiest thought" is the freely falling box, in which gravity disappears. In RG language the box is a protocol: accelerometers, clocks and test bodies inside a region of size L over a time T. The equivalence principle is the claim that this protocol has gravity in its kernel.

Theorem 2.1 (The equivalence principle is a resolution window; F01)**.** *In Fermi coordinates along a freely falling worldline, $g = \eta + O(R\,x^2)$. The connection vanishes on the worldline; the curvature does not. For a central mass the tidal tensor is* $\mathrm{diag}(-2,1,1)\,GM/r^3$*, which is traceless. The box lies in the fibre of the inertial box exactly when*

$$|R|\,L < \delta_a \qquad \text{and} \qquad |R|\,L\,T^2 < \delta_x,$$

where δ_a and δ_x are the declared differential-acceleration and displacement resolutions. In low Earth orbit $GM/r^3 = 1.28\times10^{-6}\,\mathrm{s}^{-2}$, i.e. $1284\,E$. Gradiometry at the milli-Eötvös level resolves this at signal-to-noise of order 10^5–10^6; space gradiometry of this kind has been flown [6]. Two test masses $1\,m$ apart drift $12.8\,mm$ in $100\,s$.

Corner. The equivalence principle is exact only in the infinitesimal sense: the Christoffel symbols are gauge, like RG's bisection part. Any finite box with a fine enough instrument resolves gravity, so the box is a CORRESPONDENCE only below a declared resolution.

Theorem 2.2 (The box reads energy, not structure; F02)**.** *For the linearized metric $g = -(1+2\Phi)dt^2 + (1-2\gamma\Phi)\,dx^2$:*

- *the tidal (electric) part $R_{0i0j} = \partial_i\partial_j\Phi$ is independent of γ;*
- *the spatial part R_{ijkl} is proportional to γ.*

Slow probes, clocks and accelerometers read the energy footprint N. Structure h is reached only by light, or by fast probes suppressed by v^2/c^2, exactly as in the duality theorem [2]. At the solar limb the equivalence principle alone (Einstein 1911 [3]) predicts $0.876''$; with structure (1915) the prediction is $1.751''$.

Verdict 2.3 (Floating box). The box is cornered twice. It sees gravity once the resolution is fine enough. Even a perfect box sees only half of gravity, the energy footprint. Einstein could not have derived general relativity from the box: he needed a selected field equation and light. The infinitesimal equivalence principle survives intact, as a gauge statement.

3 Arena I: protocolism against realism

Theorem 3.1 (Realism is idle beyond the quotient)**.** *By universal factorization (L2), every observable task factors through the observable quotient. Latent structure that is constant on fibres has no observable consequence. The following are of this kind:*

- *points of the manifold (F04);*
- *black-hole interiors for exterior protocols (F05);*
- *empirically equivalent interpretations of quantum mechanics (F12).*

Theorem 3.2 (Protocolism is not self-founding)**.** *Pure operationalism needs declarations that no protocol supplies:*

- *a task, since minimal panels are task-relative and no task-free equivariant selector exists (L13, F13);*

- *units, since there is no internal unit (L94, F07);*
- *an admissible class, since there is no uniform forgetting without one (L53);*
- *the hypotheses that open walls (H-reg, H-SI, and so on).*

Verdict 3.3 (Arena I). Protocolism has the upper hand on knowledge: it computes exactly what can be decided. Realism keeps the metaphysical field, but wherever it exceeds the quotient it is idle, not refuted. Bridgman's operationalism [8] needs RG's declarations to stand. The dispute becomes a ledger, not a war.

4 Arena II: RG against Einstein

Einstein's physics. General relativity wins every empirical test that RG's own gravitation stack runs: Mercury's perihelion, light deflection, the GPS clock and Pound–Rebka. The parameter γ is measured to $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [4], and the equivalence principle has been tested at $\eta \sim 10^{-15}$ [5]. RG has no rival dynamics.

Einstein's ontology: the best undermines.

	Claim of general relativity	RG corner	Witness
U1	The field equation follows from gravitation and matter	selected (Lovelock) or measured; no function of the source remainder fixes curvature	F06, P12
U2	The box (equivalence principle) grounds the theory	the box reads energy only; structure needs light	F02
U3	Spacetime points are physical	points are gauge; point coincidences are the observables (hole argument [7])	F04
U4	The metric describes the interior and the global shape	interiors are exterior correspondences; topology is a global remainder	F05, L119
U5	Matter obeys energy conditions	declared in general relativity; forced only for kinetic sources; Λ, the stiff fluid and electric tension lie outside the kinetic cone	Phys. Path P04
U6	G and M are physical constants of the geometry	geometry knows only GM/c^2; $\kappa = 8\pi G$ is a unit anchor	F07
U7	Distance is a property of spacetime	distance is protocol-relative: radar $\neq$ proper $\neq$ optical	Light L09

Einstein's quantum realism. No noncontextual value table exists: 0 of 512 assignments satisfy the Peres–Mermin square (F09). Local deterministic strategies obey $|\mathrm{CHSH}| \leq 2$ (F10). Einstein's *local* realism therefore loses. His determinism does not: a deterministic, no-signalling but nonlocal model reproduces the singlet exactly, with $|S| = 2.829$ (F10).

Verdict 4.1 (Arena II). Physics: **Einstein**. Ontology beyond the quotient (U1–U7): **RG**, which corners rather than refutes. Local realism: **against Einstein**. Determinism ("God does not play dice"): **undecided**, a CORRESPONDENCE. Einstein's own resolution of the hole argument, by point coincidences, is protocolist; in that sense the ontology corner is *Einstein against Einstein.*

5 Arena III: RG against Newton

Theorem 5.1 (Absolute space split; F03)**.**

- *Uniform velocity is in the kernel. Light clocks along and across the motion both give proper round trip* $2L/c$*, for every* v*.*
- *Rotation is in the quotient. The Sagnac delay* $4A\Omega/c^2$ *(*1.0×10^{-20} *s for a* 1 *m ring at the Earth's rate) is detected by ring lasers (L105).*

Theorem 5.2 (Newton's exact failure point; F14, imported from the duality paper)**.** *Newton's single potential is* exact *for flat structure at any field strength, with* $\Phi_{\text{kin}} = \frac{c^2}{2}(1 - N^{-2})$*. It fails exactly where structure is resolved. Structure's share of a deflection is* $\gamma v^2/c^2$*, which is* 10^{-8} *for planets and* 1 *for light. Two-speed panels refute Newton iff the relative precision is below* $\varepsilon_{\text{crit}}$*; for a slow probe plus light,* $\varepsilon_{\text{crit}} = \gamma/(2+\gamma) = 1/3$*. Newton therefore fails for light in arbitrarily weak fields, and slow probes protect him.*

Verdict 5.3 (Arena III). Absolute position and velocity: **Leibniz and RG** [9]. The bucket [10], i.e. absolute rotation: **Newton**. Gravity: **Newton** for slow probes and flat structure; **Einstein** for light and wherever structure is resolved. Newton's *hypotheses non fingo* is the most protocolist line in the history of physics.

6 Arena IV: luck

Luck is the arena in which the corner moved most. The final luck battery [16] is imported with its hashes (code SHA-256 `808f17dd...`). It contains:

- 37.7 Mbit of Tier-1 records, scored by 288 anytime tests;
- 9×10^7 synthetic Bell rounds;
- 3.4×10^7 null-path bits.

It turns the vague question "is it luck?" into three questions with three different statuses.

6.1 Luck relative to a declared gambler class

Definition 6.1 (Relative luck)**.** A record is *lucky relative to a gambler class* $\mathcal{S}$ *and information* I when no gambler in $\mathcal{S}$ that bets at fair odds using only I makes its capital cross $1/\alpha$. Each gambler's capital is a test martingale under the fair-bit null, so the test is anytime-valid by Ville's inequality [17].

Theorem 6.2 (Hardcoding: luck is class-relative; F15, F11)**.** *For every individual record* x *of length* N*, the gambler that bets on* x *gains about* $N \ln 2$*. No record is therefore lucky absolutely, and luck exists only relative to a declared class. In the final battery, three deterministic streams behave as follows:*

- *against the depth-8 context-tree gambler [18], which is finite-state, they score* -6.9*,* -7.6 *and* -7.6*:* lucky relative to that class*;*
- *against their committed oracles they score* $+725\,767$*:* unlucky relative to the declared class*.*

The three streams are a seeded coin, the digits of $\sqrt{2}$*, and a stream masked by a seeded mask. All 36 synthetic records are detected by some declared gambler.*

Proposition 6.3 (Power; F15)**.** *Non-rejection means "lucky above the detectable edge", never "lucky". At* $N = 2^{20}$ *the bias gambler has the exact power shown below. The null calibration crossed the anytime threshold on 30 of 4096 paths at diagnostic* $\alpha = 0.01$*, and the terminal threshold on 0.*

Bias	*0.001*	*0.002*	*0.003*	*0.004*	*0.005*
Power	3×10^{-5}	*0.025*	*0.53*	*0.98*	*0.99999*

Theorem 6.4 (Finite records and kernel luck; F11, L49)**.** *Without a declared class, finite records cannot separate chance from determinism: the digits of* $\sqrt{2}$ *pass the same block tests as a coin. Luck in the kernel is permanent. Two priors on an unobserved parameter stay* 1.0 *apart at every* n*, while on the observed one they merge as* $1/n$*.*

6.2 Physical luck: Bell certification and its gates

Theorem 6.5 (One-round no-signalling bound; F16)**.** *Optimize over the 24 vertices of the CHSH no-signalling polytope (16 local deterministic boxes and 8 PR boxes). The best one-round probability that an adversary guesses Alice's outcome is exactly* $3/2 - S/4$*; the battery verifies this by an independent LP to* 10^{-9}*. At* $S = 2\sqrt{2}$ *the probability is* 0.7929*. This is a single-round bound. Sequential, composable security needs the entropy accumulation theorem or quantum estimation factors (EAT/QEF) [20], which are not implemented.*

Theorem 6.6 (Each gate is a fibre; F10, F17, F18)**.** *A Bell statistic certifies the generated bits [19] only if all of the following gates hold. Each named countermodel reproduces the passing statistics while violating exactly one gate.*

Gate	*Countermodel that passes without it*
H-free (settings independent of devices)	*settings-dependent model: identical transcript, outputs guessable from* λ
H-wall (no hidden signalling)	*hidden-signalling model: identical transcript, outputs guessable*
H-transfer (test and generation rounds indistinguishable to the devices)	*test-flag leak: test win* 0.851*, every generation bit* 0
H-strat (setting-stratified estimator)	*biased settings: pooled win* 0.990*, stratified* $S = 2.000$
H-seq (sequential security: EAT/QEF)	*memory and drift models: no i.i.d. projection allowed*
H-ext (certified extractor and seed budget)	*a direct spot-flag sampler costs 7 fair bits per round plus 2 per test*
H-acq, H-custody (physical acquisition, external custody)	*synthetic transcripts are public and reproducible*
H-thm (verified quantum guessing theorem)	*imported curve, not independently verified*

In the battery, quantum, hidden-signalling and settings-dependent devices produce bit-identical *transcripts, with* $S_{\text{lower}} = 2.683$*. No statistic separates them.*

Remark 6.7 (Erratum to an earlier draft). A preliminary version of this analysis reported "certified expansion 3.5×". That figure came from an i.i.d. projection with an entropy-coded seed estimate ($N\,h(q) + 2qN$), not from an implemented sampler or a sequential proof. The final battery retracts it. The physical certificate is **not issued**, and the number of certified physical bits is 0.

Verdict 6.8 (Arena IV). Luck is cornered into three statuses.

(a) **Relative luck is lockable.** Luck relative to a declared gambler class can be detected with stated power or bounded, and is anytime-valid. It is never absolute: every record loses to its own oracle.

(b) **Physical luck is a conditional lock with named gates.** A Bell test certifies luck against every gate-respecting adversary, and only given declared luck in the settings (luck from luck). With the gates open, the certificate is *not issued*. Sample size, here 9×10^7 rounds, buys power and precision; it cannot turn a synthetic transcript into physical randomness.

(c) **Ontological luck is not definable by a protocol.** Hidden-signalling and settings-dependent deterministic models share its fibre.

Luck therefore no longer merely "neither wins nor loses". Its corner is a finite, explicit gate list, and the next step is physical: loophole-free Bell hardware with trusted settings [21].

7 Arena V: God

Here "God" means the omniscient observer of physics: Newton's sensorium, Laplace's demon [22], and Einstein's God who does not play dice. We make no theological claim.

Theorem 7.1 (The omniscience dilemma)**.** *An omniscient observer either is a protocol or is not.*

(A) ***If a protocol****, it is bound by RG's theorems:*

- *it is* contextual*: no noncontextual value table exists (Kochen–Specker [15]; 0/512, F09);*
- *it is* information-limited*: each certified chaotic step costs $\lambda/\ln 2$ bits, and a demon with B bits sees about B steps of the logistic map (horizons $7, 17, 27, 37, 47$ at $B = 10, \ldots, 50$; F08);*
- *it is* unit-conventional*: geometry knows only GM/c^2, so even complete laws fix no metre (F07);*
- *it is* task-declaring*: no task-free selector exists (L13).*

An infinite-precision demon is an infinite protocol, which is a declaration and not an observation.

(B) ***If not a protocol****, it lies outside every observation quotient. RG can neither find nor refute it.*

Verdict 7.2 (Arena V). God is cornered to the boundary of the quotient. This is as far as an observation geometry can go. Newton's sensorium, absolute space, is in the kernel (Arena III). Einstein's non-dicing God is a CORRESPONDENCE (Arena IV). Laplace's demon is a protocol with an infinite information budget. RG does not decide God; it decides what any God-protocol would have to be.

8 Scoreboard

Arena	Claim	Status	Upper hand	Witness
Box	equivalence principle for a finite box	window	resolution decides	F01
Box	the box grounds general relativity	cornered	RG (energy only)	F02
I	realism beyond the quotient	idle	protocolism (knowledge)	F04, F05, F12
I	protocolism self-founding	refuted	declarations needed	F13, F07
II	physics of gravitation	locked by data	Einstein	[4]
II	field equation forced	no-go	RG	F06
II	points, interiors, topology physical	idle	RG	F04, F05
II	local realism (QM)	refuted	against Einstein	F09, F10
II	determinism	correspondence	none	F10
III	absolute velocity	kernel	Leibniz / RG	F03
III	absolute rotation	quotient	Newton	F03
III	Newtonian gravity	exact failure point	Newton (slow, flat h) / Einstein (light)	F14
IV	chance certified by records alone	refuted	none (correspondence)	F11
IV	luck relative to a declared gambler class	lockable (with power)	class-relative	F15
IV	physical luck from Bell tests	conditional, 8 gates; not issued	gates decide	F16–F18
IV	ontological luck	not definable	none	F10, F17
IV	kernel luck forgotten	refuted	permanent residue	F11
V	noncontextual omniscience	refuted	contextual	F09
V	finite demon forecasts forever	refuted	one bit per step	F08
V	God as non-protocol	outside	undecided	—

Is there an overall winner?

- On *what observation decides*, RG holds the upper hand in every arena. This is partly by construction: RG is the arbiter of that question, so the victory is methodological, not physical.
- On *what the physics is*, general relativity beats Newton wherever structure is resolved, and RG concedes, having no rival dynamics.
- On *what exists*, nobody wins by observation. Realism is idle, not refuted; God and ontological luck lie at the boundary of the quotient.

The honest summary: *RG is the referee, not a player, except in the box, where it scores.*

9 What would move the corners

1. An RG-native protocol whose remainder *contains* the Weyl data (tidal or geodesic-deviation panels), together with a proof that its dynamics are forced. This would turn the no-go of Arena II into a derivation. It is open.

2. A functional two-speed deflection panel. This would measure structure and energy separately, as an experimental realization of the duality theorem.

3. A physical Bell deployment that closes the eight luck gates: trusted independent settings, causal isolation, test–generation indistinguishability, stratified estimation, a sequential entropy proof (EAT/QEF), a certified extractor, physical acquisition with external custody, and verified theorems. This is the only route by which the luck certificate of Arena IV can be issued.

4. Gradiometric box experiments reported as resolution windows rather than as tests of a principle.

10 Battery

`rg_corner_attacks.py` (NumPy, SciPy, SymPy, mpmath; master seed 20261026) runs 18 witnesses, F01–F18. The computed witnesses are:

- the tidal tensor and the PPN Riemann split (sympy);
- light-clock invariance and Sagnac;
- the hole-diffeomorphism invariance;
- scaling of Schwarzschild and Kepler;
- chaotic horizons at 80-digit precision;
- exhaustive Peres–Mermin and local CHSH;
- a deterministic nonlocal singlet model by Monte Carlo (4×10^5 samples);
- block-frequency tests and kernel posteriors;
- an independent 24-vertex no-signalling LP (F16);
- the test-flag-leak, biased-settings and identical-transcript Bell countermodels (F17).

F15 and F18 import the final luck battery, `rg_luck_final.py` with code SHA-256 `808f17dd...`, whose certificate is checked field by field. That battery has 27/27 integration and 37/37 unit checks, 288 recomputed gambler scores, 360 recomputed Bell chunks, and 46.7 MB of retained raw data. F05, F06 and F14 are imported, from Birkhoff, the physical path and the duality paper; their numeric witnesses are by construction. F12 is definitional. Ten canaries are rejected by computed criteria. They state that:

- the equivalence principle is exact for a finite box;
- the equivalence principle fixes light bending;
- absolute velocity is detectable;
- a demon with finite bits forecasts forever;
- a noncontextual omniscient table exists;
- a Bell violation proves indeterminism;
- finite records certify randomness;
- a finite-state test that passes certifies luck;
- a pooled CHSH win rate above $\cos^2(\pi/8)$ certifies nonlocality;
- a Bell violation on test rounds certifies the generation rounds.

Limits. The battery is not blind. Gradiometer sensitivities are order-of-magnitude, and no experimental data are reanalysed. The cornering is only as strong as the declared protocol classes.

11 Non-claims

1. General relativity is not refuted, and no rival dynamics is proposed.
2. No theological claim is made; "God" denotes the omniscient observer of physics.
3. Realism is not refuted. It is shown idle beyond the quotient.
4. The historical attributions are summaries, not exegesis.

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